The Spaghetti Engine
VISUAL SYSTEM MODELLING WORKSPACE

The Spaghetti Engine

Development Log

Fuzzy Cognitive Map modelling tool

Field	Value
Version	0.5.5 (prototype)
Log entry date	12 November 2026
Status	Released: deployed by the merge of the branch polish-0.5.4 into the GitHub main branch, together with the 0.5.5 documents and this log. Netlify deploys the site automatically.
Live URL	spaghetti-engine.netlify.app (Netlify, auto-deployed from the GitHub main branch)
Source repository	github.com/aPaarl/Spaghetti-engine, main branch
Previous version	0.5.4, 12 November 2026, pull request #11 (commit cdc194f)
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1. Purpose of This Log

This Development Log records the changes from version 0.5.4 (12 November 2026) to version 0.5.5 of The Spaghetti Engine. Version 0.5.5 follows a full check of the website: it makes the tool load faster, draws imported models without positions in a readable way, documents the Veldmeer game in detail, and records the corrections of pull request #9, which were not yet described in the documents.

The log documents what changed and why, how existing results are affected, how the changes were checked, and the deployment status.

2. Summary of Version 0.5.5

Area	Change
Faster loading	The layered layout engine (ELK) and the workbook writer (ExcelJS) load on first use. The first download falls from 3.9 MB to 1.5 MB (1.16 MB to 0.44 MB compressed).
Readable imports	A model whose concepts share one position, or have none, is spread out by module on import, with a note in the import message.
Veldmeer game	Described in the User Manual (section U.8): opened from the Help menu, eight chapters, 11 missions checked from the player's own model, quiz questions from the player's own loops and results, points, a rank and an advice note.
Corrections	Export-box chart buttons open their folded section; fixed-layout figures keep their text size in the figure editor; matrix quadrant captions keep room; convergence panels keep two columns; the Monte Carlo reading leaves drivers and fixed concepts out; analysis setups survive a tab switch; the Getting started card folds when you move on.
Wording	The workbook is described in the model menu, and the game follows the 0.5.4 interface.

3. The Changes in Detail

3.1 Faster loading

- **Code splitting.** ELK, used only by the layered Arrange layouts, and ExcelJS, used only by Export Analysis Workbook, are imported on first use. The main bundle falls from 3.9 MB to 1.5 MB, and from 1.16 MB to 0.44 MB with compression.

3.2 Importing a model

- **Spreading out.** When at least half of the concepts share one position, they are placed in a grid with one block per module (Methodological Annex, section U.5). The import message says so, and also reports concepts that had no position.

3.3 Corrections from the website check

- **Exports.** The chart buttons in the Export boxes of Transition Point and Transition Pathways open the folded section that holds the chart, so the export no longer fails silently.
- **Figure editor.** Fixed-layout figures such as the metro map no longer get enlarged text; the captions of the matrix quadrants keep room when the split lies near the edge; the convergence panels keep two columns at double-column width.
- **Monte Carlo.** The plain-language reading counts only concepts whose result can depend on the weights (Methodological Annex, section U.6).
- **Interface.** Analysis setups survive a tab switch, and the Getting started card folds when you move on instead of covering results.

4. Implementation Choices

Topic	Choice in the tool	Reason
Loading on demand	Dynamic import of ELK and ExcelJS.	Most visits use neither; the first download is less than half the size.
When to spread an import	At least half of the concepts on one position.	A hand-written file stacks every card; a model with a few shared positions is left as drawn.
Monte Carlo counts	Only concepts fed by another concept and not held fixed.	Drivers and fixed concepts cannot vary with the weights, so counting them overstated robustness.

5. Effect on Existing Results

- No analysis result changes. The Monte Carlo reading can give a less robust verdict than in 0.5.4 for the same runs, because drivers and fixed concepts are no longer counted as firm; the table of statistics is unchanged.
- Version raised from 0.5.4 to 0.5.5 in package.json and package-lock.json.

6. Defects Found and Fixed

- The corrections listed in section 3.3, found in a full check of the website after version 0.5.3.

7. Verification

- **Tests:** the automated tests pass.
- **Website:** every tab was opened on the peat meadow example at 1366 by 850 pixels with no errors and no sideways scrolling; the baseline, a comparison of two scenarios, a sensitivity sweep and 100 Monte Carlo runs

gave results; the sensitivity setup survived a tab switch, and the Monte Carlo reading counted 15 of 20 concepts, leaving out the five levers.

- **Build:** the production build completes without errors.

8. Deployment Status

- Deployed by the merge of the branch polish-0.5.4 into main, with the 0.5.5 documents and this log. Netlify rebuilds the site automatically.
- **Documents:** the 0.5.4 Word files were not available, so the 0.5.5 User Manual and Methodological Annex were rebuilt from the 0.5.3 Word files with the 0.5.4 update chapter restored from the 0.5.4 PDFs and a 0.5.5 update chapter added.
- **Build note:** npm run build fails in the local OneDrive folder because of the ampersand in the folder path; running node node_modules/vite/bin/vite.js build directly works.

9. Prototype Scope and Known Limitations

- All limitations listed in the 0.4.0 to 0.5.4 logs still apply.
- The first use of a layered layout or of the workbook export downloads its component, which takes a moment on a slow connection.

10. Related Documentation

Document	Covers
The Spaghetti Engine - User Manual (version 0.5.5)	How to use every feature; the chapters Update for Version 0.5.4 and Update for Version 0.5.5 describe the latest changes.
The Spaghetti Engine - Methodological Annex (version 0.5.5)	The methods; sections U.5 to U.7 add the spreading of imports, the Monte Carlo counting rule and loading on demand.
Development Log version 0.5.4	The 0.5.4 release: the layout for laptop screens, the Help menu, the Network tab and the journal style of the figure editor.
This Development Log	What changed in 0.5.5 and the technical work behind it.

11. Next Steps

- Integrate the update chapters into the body of the User Manual and the Methodological Annex at the next full revision.
- Wrap long concept names in figures instead of shortening them.
- Continue with the roadmap: version 0.6.0 (modelling made easier).